

ROSEBURG, OR, USA

WMO: 726904

Lat: 43.239N

Lon: 123.355W

Elev: 525

StdP: 14.42

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 24231

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.9	30.4	21.4	16.4	34.8	26.7	21.0	39.7	15.8	50.3	12.8	50.4	1.0	0	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	94.4	67.1	90.8	66.1	87.6	65.1	69.2	89.7	67.7	87.5	66.2	84.6	9.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	83.1	75.8	60.3	79.5	74.4	58.6	74.8	72.4	33.5	89.9	32.1	87.5	31.0	84.5	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.4	10.8	DB	22.7	102.4	5.7	3.7	18.7	105.0	15.3	107.2	12.2	109.3	8.1	112.0	(4)
(5)			WB	21.7	72.5	5.7	1.9	17.6	73.9	14.2	75.0	11.0	76.1	6.9	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	43.4	45.4	48.9	52.7	59.0	64.8	71.6	71.6	66.0	56.0	47.7	42.3		
	DBStd	11.77	5.70	5.07	5.50	6.06	6.40	6.23	5.32	4.81	5.60	5.56	6.04	6.32		
	HDD50	860	213	143	89	37	4	0	0	0	0	13	112	249		
	HDD65	3944	668	550	500	372	206	76	7	5	54	283	520	703		
	CDD50	2998	10	14	54	118	284	445	670	670	480	199	43	11		
	CDD65	606	0	0	0	3	21	72	212	210	83	5	0	0		
	CDH74	7371	0	0	4	98	385	935	2439	2323	1090	97	0	0		
	CDH80	3154	0	0	0	26	127	358	1130	1059	434	20	0	0		
(14) Wind	WSAvg	3.3	2.1	2.5	2.9	3.5	4.0	4.9	5.2	4.4	3.4	2.4	2.2	2.3		
(15) Precipitation	PrecAvg	33.1	5.2	3.7	3.6	2.5	1.9	0.8	0.4	0.6	1.0	2.2	5.5	5.9		
	PrecMax	60.2	11.3	9.8	7.0	7.0	6.3	2.7	3.0	3.3	3.7	5.3	15.9	15.8		
	PrecMin	21.7	0.6	1.0	0.4	0.6	0.3	0.0	0.0	0.0	0.0	0.1	1.1	0.8		
	PrecStd	7.4	2.8	2.2	1.6	1.4	1.4	0.7	0.6	0.7	0.9	1.3	3.0	3.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	64.1	73.0	83.8	90.0	95.3	100.3	99.2	94.9	82.2	69.8	62.1		
		MCWB	53.2	53.3	56.6	61.5	65.1	68.1	69.2	68.2	65.2	60.9	60.1	54.2		
	2%	DB	57.2	59.8	68.0	75.4	83.2	88.8	94.4	93.4	89.9	75.8	63.4	57.4		
		MCWB	51.4	51.5	54.1	58.1	62.4	65.9	67.4	67.6	64.6	58.7	55.1	52.4		
	5%	DB	54.5	56.6	63.5	69.8	78.1	83.7	90.4	90.2	85.1	71.2	59.5	54.4		
		MCWB	50.0	49.5	52.7	55.6	59.9	63.2	66.3	66.1	63.2	57.8	53.8	50.6		
	10%	DB	52.0	53.9	59.0	64.3	72.7	79.2	86.4	86.1	80.7	65.9	56.3	51.6		
		MCWB	48.4	48.4	50.6	53.0	57.9	61.3	64.9	65.0	61.7	56.1	51.7	48.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	55.6	58.4	63.0	67.3	70.0	72.1	71.1	68.8	63.6	60.5	56.1		
		MCDB	59.3	61.2	69.4	80.4	88.3	91.0	93.4	93.2	88.1	76.8	67.1	59.5		
	2%	WB	52.9	52.8	55.8	59.2	63.7	66.7	69.3	69.0	66.5	61.1	57.2	53.4		
		MCDB	55.9	57.6	64.5	72.8	80.6	85.9	89.7	89.7	84.4	71.8	61.4	56.2		
	5%	WB	50.7	50.8	53.6	56.5	60.9	64.4	67.7	67.5	64.7	59.1	54.6	50.9		
		MCDB	54.0	55.1	61.2	67.7	75.2	81.2	87.4	87.2	81.2	67.7	58.0	54.0		
	10%	WB	48.6	48.9	51.5	54.0	58.7	62.2	66.0	65.9	62.6	57.3	52.6	48.2		
		MCDB	51.5	53.1	57.5	62.9	70.8	76.9	84.3	83.9	77.5	64.4	55.7	50.9		
(35) Mean Daily Temperature Range	MDBR		12.0	14.6	17.7	19.7	21.5	23.0	27.1	27.7	26.4	19.8	12.5	10.5		
	5% DB	MCDBR	15.7	19.6	26.2	31.2	32.8	32.1	33.4	34.2	35.4	29.4	17.4	13.8		
		MCWBR	9.9	11.7	13.8	15.2	14.5	12.8	12.2	12.6	14.2	14.1	10.2	8.9		
	5% WB	MCDBR	13.8	16.4	23.1	28.2	29.9	29.4	31.6	31.4	30.9	23.5	14.5	12.6		
		MCWBR	9.2	10.4	12.7	14.2	13.7	12.4	12.5	12.1	13.2	12.0	9.2	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.316	0.316	0.340	0.345	0.348	0.344	0.345	0.330	0.323	0.309	0.317		
	taud		2.564	2.553	2.572	2.474	2.486	2.488	2.514	2.525	2.550	2.574	2.585	2.531		
	Ebn at Noon		265	282	296	294	294	292	292	289	287	275	263	249		
	Edh at Noon		21	25	28	33	34	34	33	31	28	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		484	739	1073	1502	1913	2153	2378	2069	1573	966	531	385		
	RadStd		66	98	122	128	163	161	119	108	118	94	62	62		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+18

Nomenclature: See separate page